Hypotheses and measurable outcomes

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Hypotheses and measurable outcomes before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

A literature comparison matrix

Without opening the old lesson, explain one key idea from A literature comparison matrix. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Methods results and limitations

Without opening the old lesson, explain one key idea from Methods results and limitations. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

A question that evidence can answer

Without opening the old lesson, explain one key idea from A question that **evidence** can answer. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Hypotheses and measurable outcomes trace and record the first step you can explain without help.

Use W2: Trace the worked case for Hypotheses and measurable outcomes. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Hypotheses and measurable outcomes explanation against the supplied facts before discussing it.

Use the worked case for Hypotheses and measurable outcomes. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Before running12 cases, you expect B support to exceed A. Write primary metric and one falsifying result.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Hypotheses and measurable outcomes facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Hypotheses and measurable outcomes facts and keep your reasoning.

After seeing faster responses, the team changes its primary claim from support improvement to speed improvement. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 Audit the limit

> Audit the limit. Apply the supplied Hypotheses and measurable outcomes facts and keep your reasoning.

A report omits this qualification: A **hypothesis** can fail honestly; changing it after results does not erase the original test. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Hypotheses and measurable outcomes facts and keep your reasoning.

Before running12 cases, you expect B support to exceed A. Write primary metric and one falsifying result.

> **My reasoning and evidence**

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Hypotheses and measurable outcomes facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Hypotheses and measurable outcomes to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled hypotheses and measurable outcomes **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of hypotheses and measurable outcomes.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Hypotheses and measurable outcomes

> Write a testable hypothesis before scoring. A team predicts that a retrieval rule will reduce unsupported answers on a fixed validation set. Explain the labelled step before checking the changed condition.

Write a testable **hypothesis** before scoring. A team predicts that a retrieval rule will reduce unsupported answers on a fixed validation set.

Ten cases are reserved. Unsupported means a factual claim absent from the approved card. **Prediction**: B has fewer unsupported cases than A.

AI ACADEMY / AI-7/7

Hypotheses and measurable outcomes

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name the directional claim

The prediction concerns unsupported cases, not writing elegance.



2. Define scoring first

Each answer either contains an unsupported claim or does not under the fixed rubric.



3. Freeze the comparison

Run both versions on the same cases after recording the hypothesis.

CHANGE ONE THING

B sounds friendlier but has the same unsupported count as A.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Name the predicted outcome and how unsupported is scored. If B sounds friendlier but ties A's unsupported count, does the recorded improvement **hypothesis** receive support?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Write a testable **hypothesis** before scoring. A team predicts that a retrieval rule will reduce unsupported answers on a fixed validation set.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Run both versions on the same cases after recording the **hypothesis**.

1. Name the directional claim

2. Freeze the comparison

3. Define scoring first

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Freeze the comparison Run both versions on the same cases after recording the **hypothesis**. A **hypothesis** makes an expected pattern explicit before seeing outcomes.

The measured definition must be clear enough for another reviewer to apply.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: B sounds friendlier but has the same unsupported count as A. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. The stated improvement **hypothesis** is not supported on these cases.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The stated improvement **hypothesis** is not supported on these cases. Friendliness is a different outcome from the predefined unsupported-case count.

Repair a missing building block

Review A literature comparison matrix (ai-7/6). Goal: Explain the **mechanism** and terminology of a literature comparison matrix.

Short practice: Explain why 8/10 indoors and 16/20 outdoors share a rate but not a controlled comparison. Which condition column must the literature matrix preserve?. Return to this week and explain what you changed.

Review Methods results and limitations (ai-7/4). Goal: Explain the **mechanism** and terminology of methods results and limitations.

Short practice: Place the five-photo procedure, 3/5 versus 4/5 finding and missing outdoor **evidence** in their proper categories. Why cannot the result establish that B is best for outdoor farming?.

Return to this week and explain what you changed.

Review A question that **evidence** can answer (ai-7/1). Goal: Explain the **mechanism** and terminology of a question that **evidence** can answer. Short practice: Which proposed diagram question has a measurable outcome?

Name its versions, six-case denominator and fixed scoring, then explain why giving only B easier tasks spoils the comparison.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

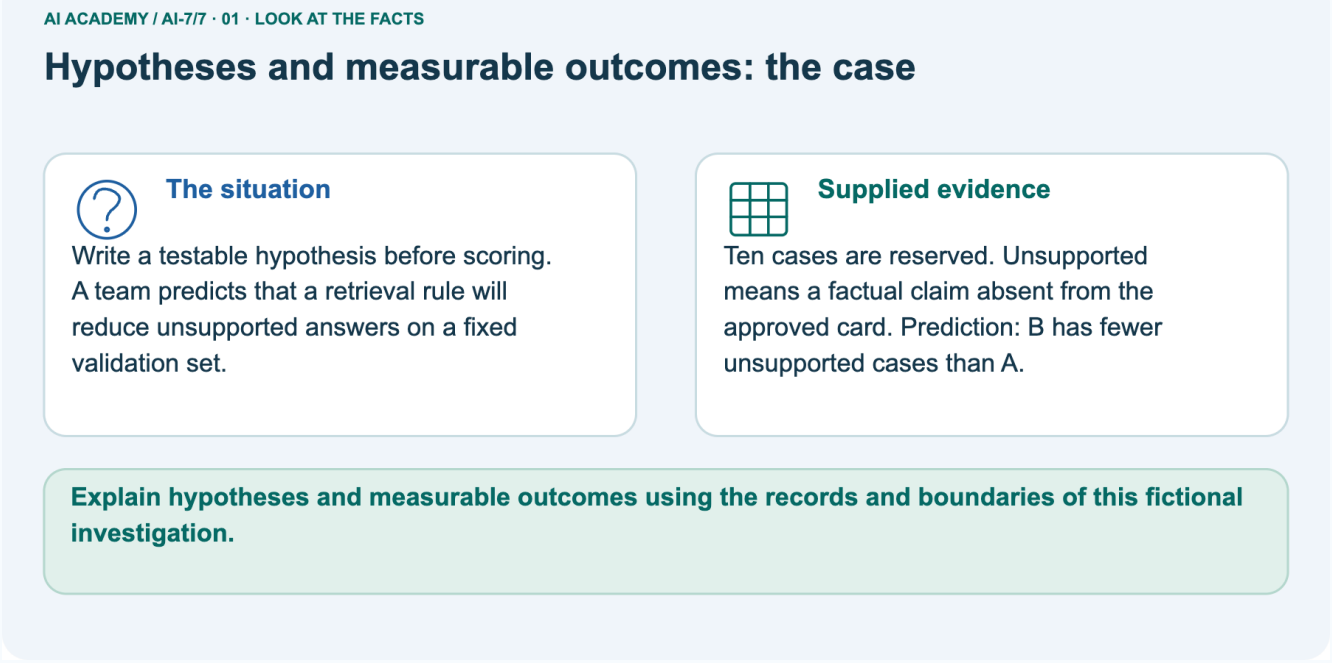
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Hypotheses and measurable outcomes picture from its labels, then record a relevant limit.

01. Hypotheses and measurable outcomes: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

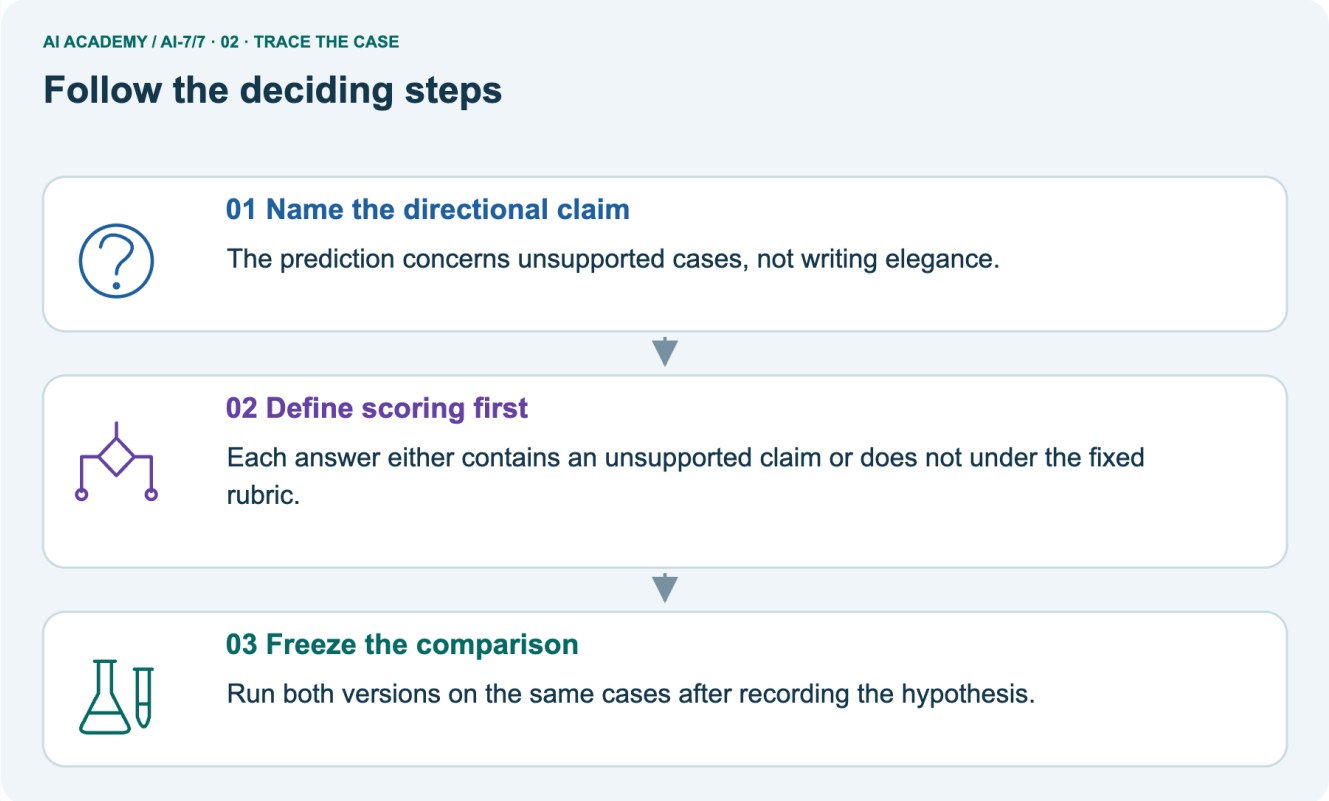
Write a testable **hypothesis** before scoring. A team predicts that a retrieval rule will reduce unsupported answers on a fixed validation set. Supplied facts: Ten cases are reserved.

Unsupported means a factual claim absent from the approved card. **Prediction**: B has fewer unsupported cases than A.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



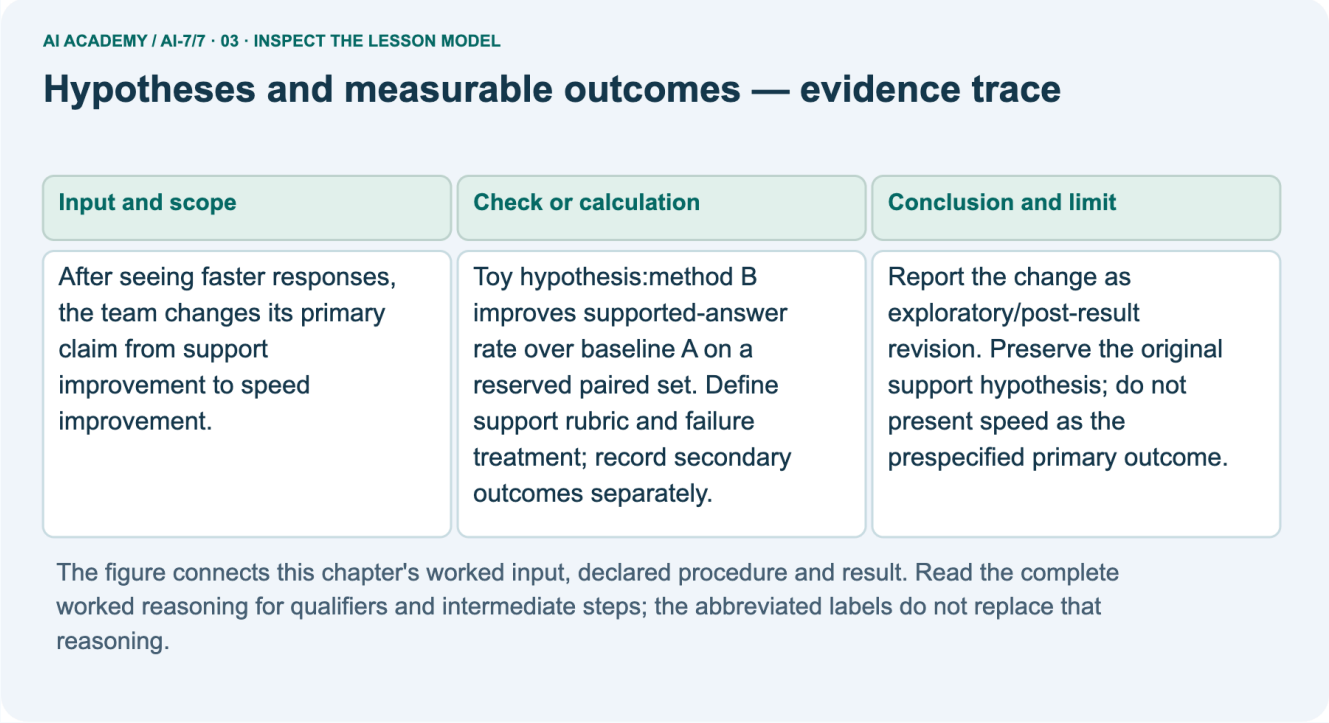
Follow the labelled picture with the matching explanation. Open the vector diagram

Name the directional claim: The **prediction** concerns unsupported cases, not writing elegance. → Define scoring first: Each answer either contains an unsupported claim or does not under the fixed rubric. → Freeze the comparison: Run both versions on the same cases after recording the **hypothesis**.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Hypotheses and measurable outcomes — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Hypotheses and measurable outcomes — **evidence** trace.

After seeing faster responses, the team changes its primary claim from support improvement to speed improvement. | Toy **hypothesis**:method B improves supported-answer rate over baseline A on a reserved paired set.

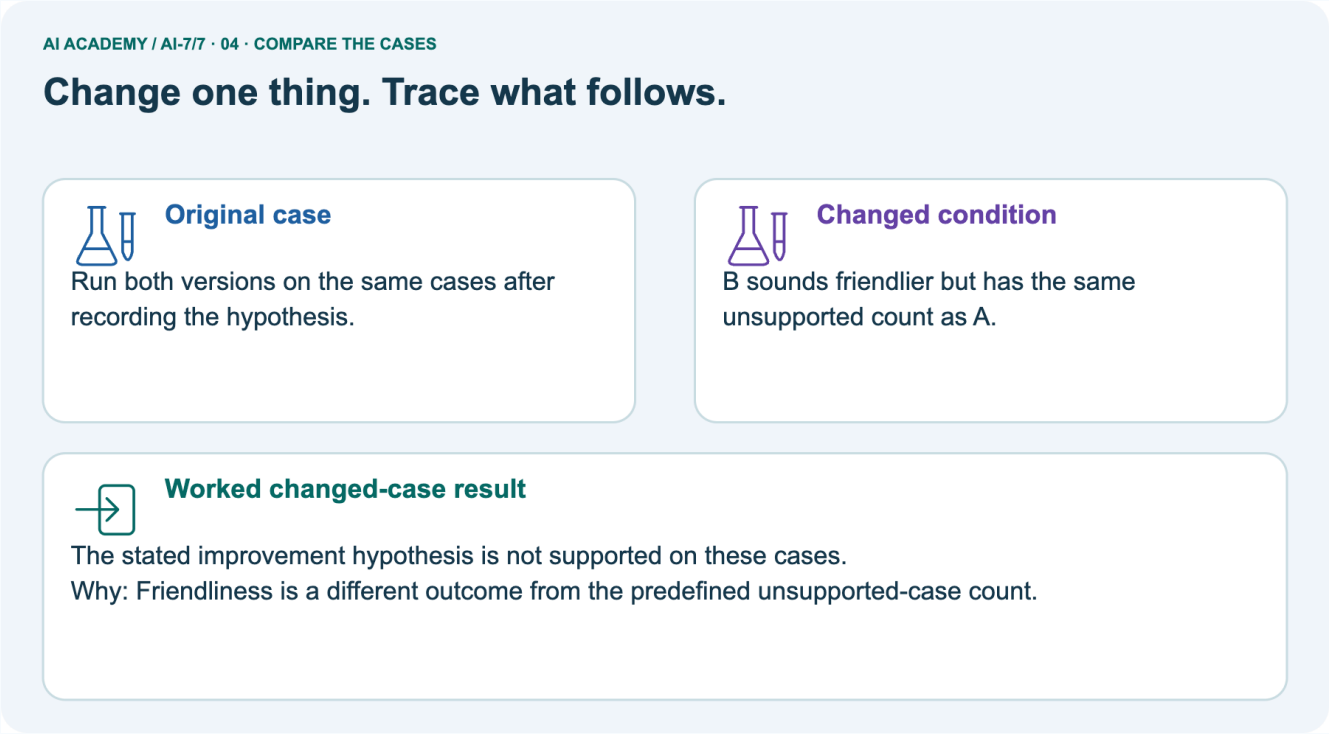
Define support rubric and failure treatment; record secondary outcomes separately. | Report the change as exploratory/post-result revision.

Preserve the original support **hypothesis**; do not present speed as the prespecified **primary outcome**.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

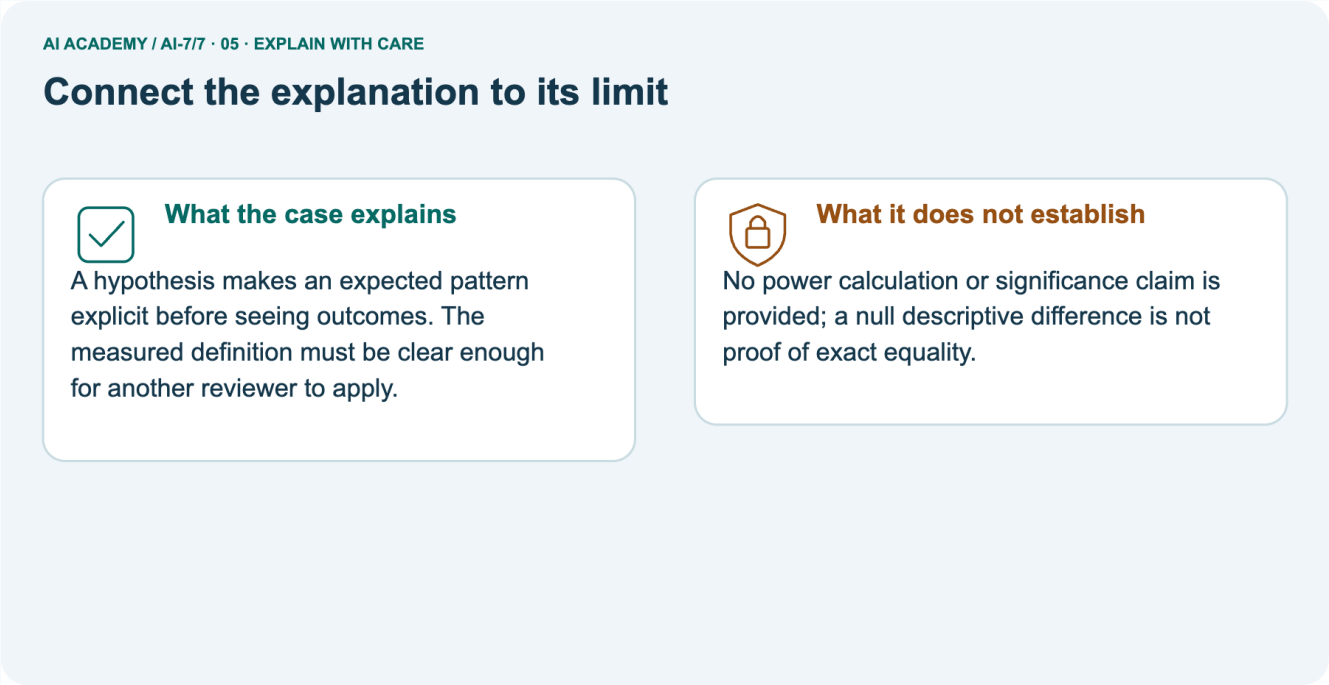
Original case: Run both versions on the same cases after recording the **hypothesis**. Changed condition: B sounds friendlier but has the same unsupported count as A.

Expected result: The stated improvement **hypothesis** is not supported on these cases. Reason: Friendliness is a different outcome from the predefined unsupported-case count.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

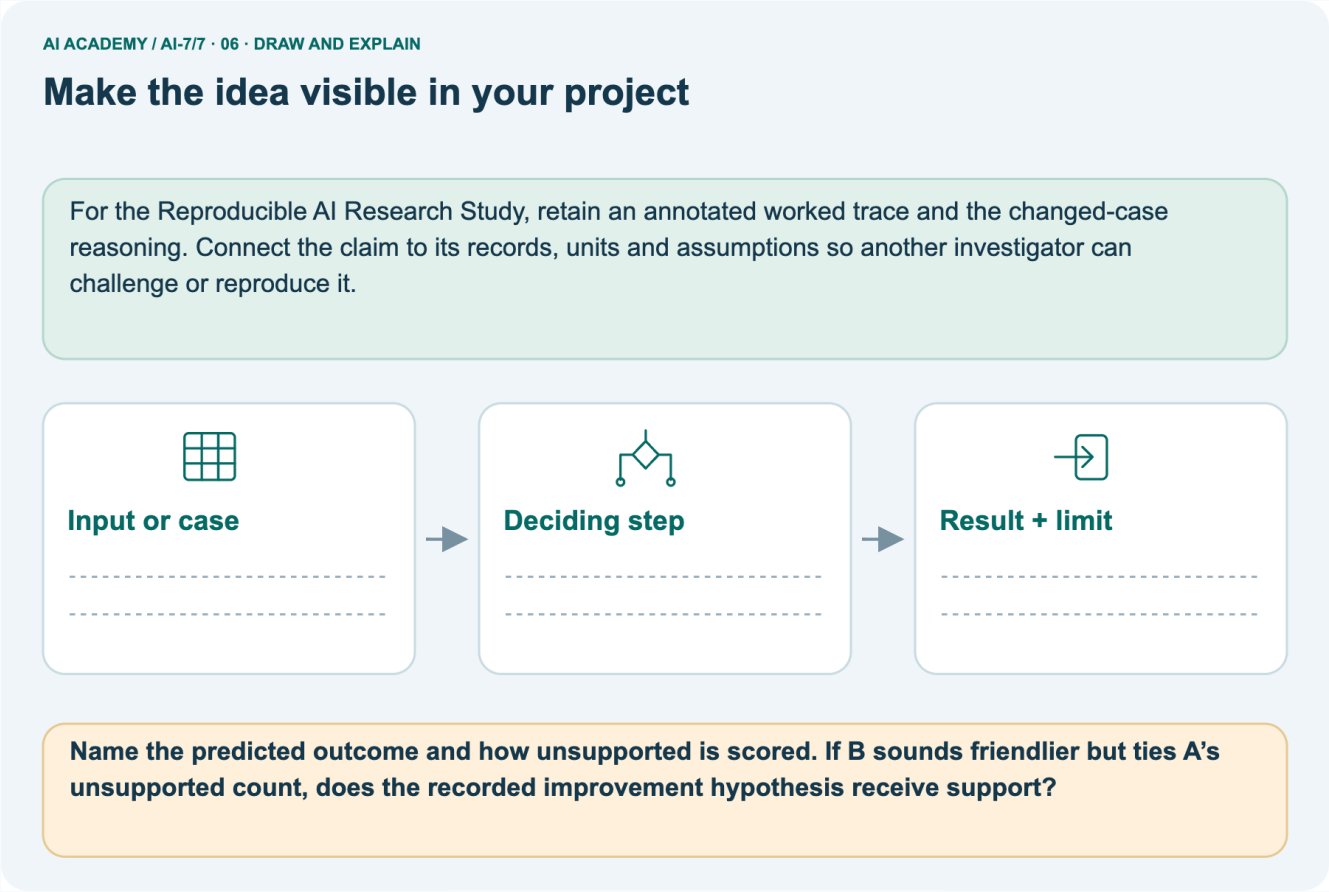
A **hypothesis** makes an expected pattern explicit before seeing outcomes. The measured definition must be clear enough for another reviewer to apply.

Limit: No power calculation or significance claim is provided; a null descriptive difference is not proof of exact equality.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Name the predicted outcome and how unsupported is scored.

If B sounds friendlier but ties A's unsupported count, does the recorded improvement **hypothesis** receive support?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.